REFERENCES CITED

Amend, J.P., LaRowe, D.E., McCollom, T.M., Shock, E.L., 2013. The energetics of organic synthesis inside and outside the cell. Philos. Trans. R. Soc. B Biol. Sci. 368, 20120255. https://doi.org/10.1098/rstb.2012.0255

Anderson, S.R., Menden-Deuer, S., 2017. Growth, Grazing, and Starvation Survival in Three Heterotrophic Dinoflagellate Species. J. Eukaryot. Microbiol. 64, 213–225. https://doi.org/https://doi.org/10.1111/jeu.12353

Ankrah, N.Y.D., May, A.L., Middleton, J.L., Jones, D.R., Hadden, M.K., Gooding, J.R., LeCleir, G.R., Wilhelm, S.W., Campagna, S.R., Buchan, A., 2014. Phage infection of an environmentally relevant marine bacterium alters host metabolism and lysate composition. ISME J. 8, 1089–1100. https://doi.org/10.1038/ismej.2013.216

Arandia-Gorostidi, N., Weber, P.K., Alonso-Sáez, L., Morán, X.A.G., Mayali, X., 2017. Elevated temperature increases carbon and nitrogen fluxes between phytoplankton and heterotrophic bacteria through physical attachment. ISME J. 11, 641–650. https://doi.org/10.1038/ismej.2016.156

Balch, W.M., Holligan, P.M., Ackleson, S.G., Voss, K.J., 1991. Biological and optical properties of mesoscale coccolithophore blooms in the Gulf of Maine. Limnol. Oceanogr. 36, 629–643. https://doi.org/https://doi.org/10.4319/lo.1991.36.4.0629

Basso, J.T.R., Jones, K.R., Jacobs, K.R., Christopher, C.J., Fielland, H.B., Campagna, S.R., Buchan, A., 2022. Growth Substrate and Prophage Induction Collectively Influence Metabolite and Lipid Profiles in a Marine Bacterium. mSystems 7, e00585-22. https://doi.org/10.1128/msystems.00585-22

Bergh, Ø., BØrsheim, K.Y., Bratbak, G., Heldal, M., 1989. High abundance of viruses found in aquatic environments. Nature 340, 467–468. https://doi.org/10.1038/340467a0

Blanco-Bercial, L., Parsons, R., Bolaños, L.M., Johnson, R., Giovannoni, S.J., Curry, R., 2022. The protist community traces seasonality and mesoscale hydrographic features in the oligotrophic Sargasso Sea . Front. Mar. Sci. .

Boenigk, J., Arndt, H., 2002. Bacterivory by heterotrophic flagellates: community structure and feeding strategies. Antonie Van Leeuwenhoek 81, 465–480. https://doi.org/10.1023/A:1020509305868

Bonnain, C., Breitbart, M., Buck, K.N., 2016. The Ferrojan Horse Hypothesis: Iron-Virus Interactions in the Ocean . Front. Mar. Sci. .

Brandes, N., Linial, M., 2019. Giant Viruses—Big Surprises. Viruses. https://doi.org/10.3390/v11050404

Bratbak, G., Egge, J.K., Heldal, M., 1993. Viral mortality of the marine alga Emiliania huxleyi (Haptophyceae) and termination of algal blooms. Mar. Ecol. Prog. Ser. 93, 39–48.

Brown, J.M., Labonté, J.M., Brown, J., Record, N.R., Poulton, N.J., Sieracki, M.E., Logares, R., Stepanauskas, R., 2020a. Single Cell Genomics Reveals Viruses Consumed by Marine Protists. Front. Microbiol.

Brown, J.M., Labonté, J.M., Brown, J., Record, N.R., Poulton, N.J., Sieracki, M.E., Logares, R., Stepanauskas, R., 2020b. Single Cell Genomics Reveals Viruses Consumed by Marine Protists . Front. Microbiol. .

Brussaard, C., Martínez Martínez, J., 2008. Algal bloom viruses. Plant Viruses 2, 1–13.

Brussaard, C.P., 2004. Optimization of Procedures for Counting Viruses by Flow Cytometry. Appl. Environ. Microbiol. 70, 1506–1513. https://doi.org/10.1128/AEM.70.3.1506-1513.2004

Brussaard, C.P., Kempers, R.S., Kop, A.J., Riegman, R., Heldal, M., 1996. Virus-like particles in a summer bloom of Emiliania huxleyi in the North Sea. Aquat. Microb. Ecol. 10, 105–113.

Buchan, A., González José, M., Moran, M.A., 2005. Overview of the Marine Roseobacter Lineage. Appl. Environ. Microbiol. 71, 5665–5677. https://doi.org/10.1128/AEM.71.10.5665-5677.2005

Cael, B.B., Bisson, K., Conte, M., Duret, M.T., Follett, C.L., Henson, S.A., Honda, M.C., Iversen, M.H., Karl, D.M., Lampitt, R.S., Mouw, C.B., Muller-Karger, F., Pebody, C.A., Smith, K.L., Talmy, D., 2021. Open ocean particle flux variability from surface to seafloor. Geophys. Res. Lett. 48, 1–10. https://doi.org/10.1029/2021gl092895

Calfee, B.C., Glasgo, L.D., Zinser, E.R., 2022. Prochlorococcus Exudate Stimulates Heterotrophic Bacterial Competition with Rival Phytoplankton for Available Nitrogen. MBio. https://doi.org/10.1128/MBIO.02571-21

Choi, K.-H., Yang, E.J., Kim, D., Kang, H.-K., Noh, J.H., Kim, C.-H., 2012. The influence of coastal waters on distributions of heterotrophic protists in the northern East China Sea, and the impact of protist grazing on phytoplankton. J. Plankton Res. 34, 886–904. https://doi.org/10.1093/plankt/fbs046

Christaki, U., Courties, C., Massana, R., Catala, P., Lebaron, P., Gasol, J.M., Zubkov, M. V, 2011. Optimized routine flow cytometric enumeration of heterotrophic flagellates using SYBR Green I. Limnol. Oceanogr. Methods 9, 329–339. https://doi.org/https://doi.org/10.4319/lom.2011.9.329

Corinaldesi, C., Dell'Anno, A., Magagnini, M., Danovaro, R., 2010. Viral decay and viral production rates in continental-shelf and deep-sea sediments of the Mediterranean Sea. FEMS Microbiol. Ecol. 72, 208–218. https://doi.org/10.1111/j.1574-6941.2010.00840.x

De Corte, D., Martínez, J.M., Cretoiu, M.S., Takaki, Y., Nunoura, T., Sintes, E., Herndl, G.J., Yokokawa, T., 2019. Viral Communities in the Global Deep Ocean Conveyor Belt Assessed by Targeted Viromics. Front. Microbiol. 10.

De Paepe, M., Taddei, F., 2006. Viruses' Life History: Towards a Mechanistic Basis of a Trade-Off between Survival and Reproduction among Phages. PLOS Biol. 4, e193.

Dekas, A.E., Parada, A.E., Mayali, X., Fuhrman, J.A., Wollard, J., Weber, P.K., Pett-Ridge, J., 2019. Characterizing Chemoautotrophy and Heterotrophy in Marine Archaea and Bacteria With Single-Cell Multi-isotope NanoSIP . Front. Microbiol. .

DeLong, J.P., Van Etten, J.L., Al-Ameeli, Z., Agarkova, I. V, Dunigan, D.D., 2023. The consumption of viruses returns energy to food chains. Proc. Natl. Acad. Sci. 120, e2215000120. https://doi.org/10.1073/pnas.2215000120

Deng, L., Krauss, S., Feichtmayer, J., Hofmann, R., Arndt, H., Griebler, C., 2014. Grazing of heterotrophic flagellates on viruses is driven by feeding behaviour. Environ. Microbiol. Rep. 6, 325–330. https://doi.org/https://doi.org/10.1111/1758-2229.12119

Dong, R., Goodbrake, C., Harrington, H.A., Pogudin, G., 2021. Differential elimination for dynamical models via projections with applications to structural identifiability. SIAM J. Appl. Algebr. Geom. 7, 194–235.

Dutkiewicz, S., Cermeno, P., Jahn, O., Follows, M.J., Hickman, A.E., Taniguchi, D.A.A., Ward, B.A., 2020. Dimensions of marine phytoplankton diversity. Biogeosciences 17, 609–634. https://doi.org/10.5194/bg-17-609-2020

Dutkiewicz, S., Follows, M., Bragg, J., 2009. Modeling the coupling of ocean ecology and biogeochemistry. Glob. Biogeochem. Cycles - Glob. BIOGEOCHEM CYCLE 23. https://doi.org/10.1029/2008GB003405

Edwards, K.F., Steward, G.F., 2018. Host Traits Drive Viral Life Histories across Phytoplankton Viruses. Am. Nat. 191, 566–581. https://doi.org/10.1086/696849

Edwards, K.F., Steward, G.F., Schvarcz, C.R., 2021. Making sense of virus size and the tradeoffs shaping viral fitness. Ecol. Lett. 24, 363–373. https://doi.org/https://doi.org/10.1111/ele.13630

Elveborg, S., Monteil, V.M., Mirazimi, A., 2022. Methods of Inactivation of Highly Pathogenic Viruses for Molecular, Serology or Vaccine Development Purposes. Pathogens. https://doi.org/10.3390/pathogens11020271

Farzad, R., Ha, A.D., Aylward, F.O., 2022. Diversity and genomics of giant viruses in the North Pacific Subtropical Gyre . Front. Microbiol. .

Fileman, E.S., Cummings, D.G., Llewellyn, C.A., 2002. Microplankton community structure and the impact of microzooplankton grazing during an Emiliania huxleyi bloom, off the Devon coast. J. Mar. Biol. Assoc. United Kingdom 82, 359–368. https://doi.org/DOI: 10.1017/S0025315402005593

Follows, M.J., Dutkiewicz, S., Grant, S., Chisholm, S.W., 2007. Emergent Biogeography of Microbial Communities in a Model Ocean. Science (80-.). 315, 1843–1846. https://doi.org/10.1126/science.1138544

Fuhrman, J.A., 1999. Marine viruses and their biogeochemical and ecological effects. Nature 399, 541–548. https://doi.org/10.1038/21119

Garcia, N.S., Talmy, D., Fu, W. -W., Larkin, A.A., Lee, J., Martiny, A.C., 2022. The diel cycle of surface ocean elemental stoichiometry has implications for ocean productivity. Global Biogeochem. Cycles 1–13. https://doi.org//2021gb007092

Gilg, I.C., Archer, S.D., Floge, S.A., M, F.D., Vermont, A.I., Leavitt, A.H., Wilson, W.H., Martínez Martínez, J., 2016. Differential gene expression is tied to photochemical efficiency reduction in virally infected Emiliania huxleyi. Mar. Ecol. Prog. Ser. 555, 13–27.

Glasgo, L.D., Lukasiak, K.L., Zinser, E.R., Glasgo, L.D., Lukasiak, K.L., Zinser, E.R., 2024. Expanding the capabilities of MuGENT for large-scale genetic engineering of the fastest-replicating species, Vibrio natriegens. Microbiol. Spectr. 12, e0396423.

Gonzalez, J., Suttle, C., 1993. Grazing by marine nanoflagellates on viruses and virus-sized particles: Ingestion and digestion. Mar. Ecol. Ser. - MAR ECOL-PROGR SER 94, 1–10. https://doi.org/10.3354/meps094001

Goode, A.G., Fields, D.M., Archer, S.D., Martínez Martínez, J., 2019. Physiological responses of Oxyrrhis marina to a diet of virally infected Emiliania huxleyi. PeerJ 7, e6722. https://doi.org/10.7717/peerj.6722

Guidi, L., Chaffron, S., Bittner, L., Eveillard, D., Larhlimi, A., Roux, S., Darzi, Y., Audic, S., Berline, L., Brum, J.R., Coelho, L.P., Espinoza, J.C.I., Malviya, S., Sunagawa, S., Dimier, C., Kandels-Lewis, S., Picheral, M., Poulain, J., Searson, S., Stemmann, L., Not, F., Hingamp, P., Speich, S., Follows, M., Karp-Boss, L., Boss, E., Ogata, H., Pesant, S., Weissenbach, J., Wincker, P., Acinas, S.G., Bork, P., de Vargas, C., Iudicone, D., Sullivan, M.B., Raes, J., Karsenti, E., Bowler, C., Gorsky, G., Coordinators, T.O.C., 2016. Plankton networks driving carbon export in the oligotrophic ocean. Nature 532, 465–470. https://doi.org/10.1038/nature16942

Guillard, R.R.L., 1975. Culture of phytoplankton for feeding marine invertebrates., in: Smith, W.L., Chanley, M.H. (Eds.), Culture of Marine Invertebrate Animals. Plenum Press, New York, USA, pp. 26–60.

Hadas, E., Marie, D., Shpigel, M., Ilan, M., 2006. Virus predation by sponges is a new nutrient-flow pathway in coral reef food webs. Limnol. Oceanogr. - LIMNOL Ocean. 51, 1548–1550. https://doi.org/10.4319/lo.2006.51.3.1548

Haidar, A.T., Thierstein, H.R., 2001. Coccolithophore dynamics off Bermuda (N. Atlantic). Deep Sea Res. Part II Top. Stud. Oceanogr. 48, 1925–1956. https://doi.org/https://doi.org/10.1016/S0967-0645(00)00169-7

Halewood, E., Opalk, K., Custals, L., Carey, M., Hansell, D.A., Carlson, C.A., 2022. Determination of dissolved organic carbon and total dissolved nitrogen in seawater using High Temperature Combustion Analysis . Front. Mar. Sci. .

Harvey, E.L., Bidle, K.D., Johnson, M.D., 2015. Consequences of strain variability and calcification in Emiliania huxleyi on microzooplankton grazing. J. Plankton Res. 37, 1137–1148. https://doi.org/10.1093/plankt/fbv081

Hawco, N.J., Tagliabue, A., Twining, B.S., 2022. Manganese Limitation of Phytoplankton Physiology and Productivity in the Southern Ocean. Global Biogeochem. Cycles 36, e2022GB007382. https://doi.org/https://doi.org/10.1029/2022GB007382

Hennemuth, W., Rhoads, L.S., Eichelberger, H., Watanabe, M., Van Bell, K.M., Ke, L., Kim, H., Nguyen, G., Jonas, J.D., Veith, D., Van Bell, C.T., 2008. Ingestion and Inactivation of Bacteriophages by Tetrahymena. J. Eukaryot. Microbiol. 55, 44–50. https://doi.org/https://doi.org/10.1111/j.1550-7408.2007.00303.x

Highfield, A., Evans, C., Walne, A., Miller, P.I., Schroeder, D.C., 2014. How many Coccolithovirus genotypes does it take to terminate an Emiliania huxleyi bloom? Virology 466–467, 138–145. https://doi.org/https://doi.org/10.1016/j.virol.2014.07.017

Hinson, A., Papoulis, S., Fiet, L., Knight, M., Cho, P., Szeltner, B., Sgouralis, I., Talmy, D., 2023a. A model of algal-virus population dynamics reveals underlying controls on material transfer. Limnol. Oceanogr. 68, 165–180. https://doi.org/https://doi.org/10.1002/lno.12256

Hinson, A., Papoulis, S., Fiet, L., Knight, M., Cho, P., Szeltner, B., Sgouralis, I., Talmy, D., 2023b. A model of algal-virus population dynamics reveals underlying controls on material transfer. Limnol. Oceanogr. 68, 165–180. https://doi.org/10.1002/lno.12256

Holling, C.S., 1959. The Components of Predation as Revealed by a Study of Small-Mammal Predation of the European Pine Sawfly. Can. Entomol. 91, 293–320. https://doi.org/DOI: 10.4039/Ent91293-5

Inomura, K., Deutsch, C., Jahn, O., Dutkiewicz, S., Follows, M.J., 2022. Global patterns in marine organic matter stoichiometry driven by phytoplankton ecophysiology. Nat. Geosci. 15, 1034–1040. https://doi.org/10.1038/s41561-022-01066-2

Jover, L.F., Effler, T.C., Buchan, A., Wilhelm, S.W., Weitz, J.S., 2014. The elemental composition of virus particles: implications for marine biogeochemical cycles. Nat. Rev. Microbiol. 12, 519–528. https://doi.org/10.1038/nrmicro3289

Kaneko, H., Blanc-Mathieu, R., Endo, H., Chaffron, S., Delmont, T.O., Gaia, M., Henry, N., Hernández-Velázquez, R., Nguyen, C.H., Mamitsuka, H., Forterre, P., Jaillon, O., de Vargas, C., Sullivan, M.B., Suttle, C.A., Guidi, L., Ogata, H., 2021. Eukaryotic virus composition can predict the efficiency of carbon export in the global ocean. iScience 24, 102002. https://doi.org/https://doi.org/10.1016/j.isci.2020.102002

Kepner Jr., R.L., Wharton Jr., R.A., Suttle, C.A., 1998. Viruses in Antarctic lakes. Limnol. Oceanogr. 43, 1754–1761. https://doi.org/https://doi.org/10.4319/lo.1998.43.7.1754

Klose, T., Rossmann, M.G., 2014. Structure of large dsDNA viruses. Biological Chemistry 395, 711–719. https://doi.org/doi:10.1515/hsz-2014-0145

Lara, E., Vaqué, D., Sà, E.L., Boras, J.A., Gomes, A., Borrull, E., Díez-Vives, C., Teira, E., Pernice, M.C., Garcia, F.C., Forn, I., Castillo, Y.M., Peiró, A., Salazar, G., Morán, X.A.G., Massana, R., Catalá, T.S., Luna, G.M., Agustí, S., Estrada, M., Gasol, J.M., Duarte, C.M., 2017. Unveiling the role and life strategies of viruses from the surface to the dark ocean. Sci. Adv. 3, e1602565. https://doi.org/10.1126/sciadv.1602565

Lawrence, J., Töpper, J., Petelenz-Kurdziel, E., Bratbak, G., Larsen, A., Thompson, E., Troedsson, C., Ray, J.L., 2018. Viruses on the menu: The appendicularian Oikopleura dioica efficiently removes viruses from seawater. Limnol. Oceanogr. 63, S244–S253. https://doi.org/https://doi.org/10.1002/lno.10734

Lennon, J.T., Abramoff, R.Z., Allison, S.D., Burckhardt, R.M., Deangelis, K.M., Dunne, J.P., Frey, S.D., Friedlingstein, P., Hawkes, C. V, Hungate, B.A., Khurana, S., Kivlin, S.N., Levine, N.M., Manzoni, S., Martiny, A.C., Martiny, J.B.H., Nguyen, N.K., Rawat, M., Talmy, D., Vogt, M., Wieder, W.R., Zakem, E.J., Lennon, J.T., Abramoff, R.Z., Allison, S.D., Burckhardt, R.M., Deangelis, K.M., Dunne, J.P., Frey, S.D., Friedlingstein, P., Hawkes, C. V, Hungate, B.A., Khurana, S., Kivlin, S.N., Levine, N.M., Manzoni, S., Martiny, A.C., Martiny, J.B.H., Nguyen, N.K., Rawat, M., Talmy, D., Vogt, M., Wieder, W.R., Zakem, E.J., 2024. Priorities, opportunities, and challenges for integrating microorganisms into Earth system models for climate change prediction. MBio 1–9.

Lin, Y.-C., Chin, C.-P., Yang, J.W., Chiang, K.-P., Hsieh, C., Gong, G.-C., Shih, C.-Y., Chen, S.-Y., 2022. How Communities of Marine Stramenopiles Varied with Environmental and Biological Variables in the Subtropical Northwestern Pacific Ocean. Microb. Ecol. 83, 916–928. https://doi.org/10.1007/s00248-021-01788-7

Maat, D.S., Crawfurd, K.J., Timmermans, K.R., Brussaard, C.P.D., 2014. Elevated CO2 and Phosphate Limitation Favor Micromonas pusilla through Stimulated Growth and Reduced Viral Impact. Appl. Environ. Microbiol. 80, 3119–3127. https://doi.org/10.1128/AEM.03639-13

Mackinder, L.C.M., Worthy, C.A., Biggi, G., Hall, M., Ryan, K.P., Varsani, A., Harper, G.M., Wilson, W.H., Brownlee, C., Schroeder, D.C., 2009. A unicellular algal virus, Emiliania huxleyi virus 86, exploits an animal-like infection strategy. J. Gen. Virol. 90, 2306–2316. https://doi.org/10.1099/vir.0.011635-0

Marie, D., Partensky, F., Vaulot, D., Brussaard, C., 1999. Enumeration of Phytoplankton, Bacteria, and Viruses in Marine Samples. Curr. Protoc. Cytom. 10, 11.11.1-11.11.15. https://doi.org/https://doi.org/10.1002/0471142956.cy1111s10

Martínez, J.M., Martinez-Hernandez, F., Martinez-Garcia, M., 2020. Single-virus genomics and beyond. Nat. Rev. Microbiol. 1–12.

Martínez Martínez, J., Norland, S., Thingstad, T.F., Schroeder, D.C., Bratbak, G., Wilson, W.H., Larsen, A., 2006. Variability in microbial population dynamics between similarly perturbed mesocosms. J. Plankton Res. 28, 783–791. https://doi.org/10.1093/plankt/fbl010

Martínez Martínez, J., Schroeder, D.C., Wilson, W.H., 2012. Dynamics and genotypic composition of Emiliania huxleyi and their co-occurring viruses during a coccolithophore bloom in the North Sea. FEMS Microbiol. Ecol. 81, 315–323. https://doi.org/10.1111/j.1574-6941.2012.01349.x

Martiny, A.C., Hagstrom, G.I., DeVries, T., Letscher, R.T., Britten, G.L., Garcia, C.A., Galbraith, E., Karl, D., Levin, S.A., Lomas, M.W., Moreno, A.R., Talmy, D., Wang, W., Matsumoto, K., 2022. Marine phytoplankton resilience may moderate oligotrophic ecosystem responses and biogeochemical feedbacks to climate change. Limnol. Oceanogr. 2, 378–389. https://doi.org/10.1002/lno.12029

Mayali, X., Weber, P.K., 2018. Quantitative isotope incorporation reveals substrate partitioning in a coastal microbial community. FEMS Microbiol. Ecol. 94, fiy047. https://doi.org/10.1093/femsec/fiy047

Mayali, X., Weber, P.K., Brodie, E.L., Mabery, S., Hoeprich, P.D., Pett-Ridge, J., 2012. High-throughput isotopic analysis of RNA microarrays to quantify microbial resource use. ISME J. 6, 1210–1221. https://doi.org/10.1038/ismej.2011.175

Mayali, X., Weber, P.K., Mabery, S., Pett-Ridge, J., 2014. Phylogenetic Patterns in the Microbial Response to Resource Availability: Amino Acid Incorporation in San Francisco Bay. PLoS One 9, e95842.

Mayers, K.M.J., Kuhlisch, C., Basso, J.T.R., Saltvedt, M.R., Buchan, A., Sandaa, R.-A., 2023. Grazing on Marine Viruses and Its Biogeochemical Implications. MBio 14, e01921-21. https://doi.org/10.1128/mbio.01921-21

Mayers, K.M.J., Lawrence, J., Sandnes Skaar, K., Töpper, J.P., Petelenz, E., Rydningen Saltvedt, M., Sandaa, R.-A., Larsen, A., Bratbak, G., Ray, J.L., 2021. Removal of large viruses and their dispersal through fecal pellets of the appendicularian Oikopleura dioica during Emiliania huxleyi bloom conditions. Limnol. Oceanogr. 66, 3963–3975. https://doi.org/https://doi.org/10.1002/lno.11935

Menden-Deuer, S., Lessard, E.J., 2000. Carbon to volume relationships for dinoflagellates, diatoms, and other protist plankton. Limnol. Oceanogr. 45, 569–579. https://doi.org/https://doi.org/10.4319/lo.2000.45.3.0569

Metropolis, N., Rosenbluth, A.W., Rosenbluth, M.N., Teller, A.H., Teller, E., 1953. Equation of State Calculations by Fast Computing Machines. J. Chem. Phys. 21, 1087–1092. https://doi.org/10.1063/1.1699114

Moreira, D., Brochier-Armanet, C., 2008. Giant viruses, giant chimeras: The multiple evolutionary histories of Mimivirus genes. BMC Evol. Biol. 8, 12. https://doi.org/10.1186/1471-2148-8-12

Nissimov, J.I., Talmy, D., Haramaty, L., Fredricks, H.F., Zelzion, E., Knowles, B., Eren, A.M., Vandzura, R., Laber, C.P., Schieler, B.M., Johns, C.T., More, K.D., Coolen, M.J.L., Follows, M.J., Bhattacharya, D., Van Mooy, B.A.S., Bidle, K.D., 2019. Biochemical diversity of glycosphingolipid biosynthesis as a driver of Coccolithovirus competitive ecology. Environ. Microbiol. 21, 2182–2197. https://doi.org/https://doi.org/10.1111/1462-2920.14633

Noble, R.T., Fuhrman, J.A., 1999. Breakdown and microbial uptake of marine viruses and other lysis products. Aquat. Microb. Ecol. 20, 1–11.

Noble, R.T., Fuhrman, J.A., 1997. Virus decay and its causes in coastal waters. Appl. Environ. Microbiol. 63, 77–83. https://doi.org/10.1128/aem.63.1.77-83.1997

Olive, M., Moerman, F., Fernandez-Cassi, X., Altermatt, F., Kohn, T., 2022. Removal of

Waterborne Viruses by Tetrahymena pyriformis Is Virus-Specific and Coincides with Changes in Protist Swimming Speed. Environ. Sci. Technol. 56, 4062–4070. https://doi.org/10.1021/acs.est.1c05518

Olson, M.B., Strom, S.L., 2002. Phytoplankton growth, microzooplankton herbivory and community structure in the southeast Bering Sea: insight into the formation and temporal persistence of an Emiliania huxleyi bloom. Deep Sea Res. Part II Top. Stud. Oceanogr. 49, 5969–5990. https://doi.org/https://doi.org/10.1016/S0967-0645(02)00329-6

Omta, A.W., Heiny, E.A., Rajakaruna, H., Talmy, D., Follows, M.J., 2023. Trophic model closure influences ecosystem response to enrichment. Ecol. Modell. 475, 110183. https://doi.org/10.1016/j.ecolmodel.2022.110183

Omta, A.W., Talmy, D., Inomura, K., Irwin, A.J., Finkel, Z. V, Sher, D., Liefer, J.D., Follows, M.J., 2020. Quantifying nutrient throughput and DOM production by algae in continuous culture. J. Theor. Biol. 494, 110214. https://doi.org/https://doi.org/10.1016/j.jtbi.2020.110214

Pagarete, A., Allen, M.J., Wilson, W.H., Kimmance, S.A., De Vargas, C., 2009. Host–virus shift of the sphingolipid pathway along an Emiliania huxleyi bloom: survival of the fattest. Environ. Microbiol. 11, 2840–2848. https://doi.org/https://doi.org/10.1111/j.1462-2920.2009.02006.x

Papoulis, S.E., Wilhelm, S.W., Talmy, D., Zinser, E.R., 2021. Nutrient Loading and Viral Memory Drive Accumulation of Restriction Modi fi cation Systems in Bloom-Forming. MBio 12, 1–18.

Peduzzi, P., Weinbauer, M.G., 1993. Effect of concentrating the virus-rich 2-2nm size fraction of seawater on the formation of algal flocs (marine snow). Limnol. Oceanogr. 38, 1562–1565. https://doi.org/https://doi.org/10.4319/lo.1993.38.7.1562

preprint-Martínez Martínez, J., Talmy, D., Kimbrel, J.A., Weber, P.K., Mayali, X., 2024. Coastal bacteria and protists assimilate viral carbon and nitrogen. bioRxiv 2024.06.14.598912. https://doi.org/10.1101/2024.06.14.598912

Proctor, L.M., Fuhrman, J.A., 1991. Roles of viral infection in organic particle flux. Mar. Ecol. Prog. Ser. 69, 133–142.

Rajakaruna, H., Omta, A.W., Carr, E., Talmy, D., 2022. Linear scaling between microbial predator and prey densities in the global ocean. Environ. Microbiol. 25, 1–9. https://doi.org/10.1111/1462-2920.16274

Rose, J.M., Caron, D.A., Sieracki, M.E., 2004. Counting heterotrophic nanoplanktonic protists in cultures and aquatic communities by flow cytometry . Aquat. Microb. Ecol. 34, 263–277.

Sanders, R.W., Caron, D.A., Berninger, U.-G., 1992. Relationships between bacteria and heterotrophic nanoplankton in marine and fresh waters: an inter-ecosystem comparison. Mar. Ecol. Prog. Ser. 86, 1–14.

Schatz, D., Rosenwasser, S., Malitsky, S., Wolf, S.G., Feldmesser, E., Vardi, A., 2017. Communication via extracellular vesicles enhances viral infection of a cosmopolitan alga. Nat. Microbiol. 2, 1485–1492. https://doi.org/10.1038/s41564-017-0024-3

Schulz, F., Roux, S., Paez-Espino, D., Jungbluth, S., Walsh, D.A., Denef, V.J., McMahon, K.D., Konstantinidis, K.T., Eloe-Fadrosh, E.A., Kyrpides, N.C., Woyke, T., 2020. Giant virus diversity and host interactions through global metagenomics. Nature 578, 432–436. https://doi.org/10.1038/s41586-020-1957-x

Sheyn, U., Rosenwasser, S., Lehahn, Y., Barak-Gavish, N., Rotkopf, R., Bidle, K.D., Koren, I., Schatz, D., Vardi, A., 2018. Expression profiling of host and virus during a coccolithophore bloom provides insights into the role of viral infection in promoting carbon export. ISME J. 12, 704–713. https://doi.org/10.1038/s41396-017-0004-x

Stock, C.A., Dunne, J.P., John, J.G., 2014. Global-scale carbon and energy flows through the marine planktonic food web: An analysis with a coupled physical–biological model. Prog. Oceanogr. 120, 1–28. https://doi.org/https://doi.org/10.1016/j.pocean.2013.07.001

Suttle, C.A., 2005. Viruses in the sea. Nature 437, 356–361. https://doi.org/10.1038/nature04160

Suttle, C.A., 1994. The significance of viruses to mortality in aquatic microbial communities. Microb. Ecol. 28, 237–243. https://doi.org/10.1007/BF00166813

Suttle, C.A., Feng, C., 1992. Mechanisms and Rates of Decay of Marine Viruses in Seawater. Appl. Environ. Microbiol. 58, 3721–3729. https://doi.org/10.1128/aem.58.11.3721-3729.1992

Talmy, D., Carr, E., Rajakaruna, H., Våge, S., Omta, A.W., 2024. Killing the predator: impacts of highest-predator mortality on the global-ocean ecosystem structure. Biogeosciences 21, 2493–2507. https://doi.org/10.5194/bg-21-2493-2024

Thamatrakoln, K., Talmy, D., Haramaty, L., Maniscalco, C., Latham, J.R., Knowles, B., Natale, F., Coolen, M.J.L., Follows, M.J., Bidle, K.D., 2019. Light regulation of coccolithophore host–virus interactions. New Phytol. 221, 1289–1302. https://doi.org/10.1111/nph.15459

Thomassen, E., Gielen, G., Schütz, M., Schoehn, G., Abrahams, J.P., Miller, S., van Raaij, M.J., 2003. The Structure of the Receptor-binding Domain of the Bacteriophage T4 Short Tail Fibre Reveals a Knitted Trimeric Metal-binding Fold. J. Mol. Biol. 331, 361–373. https://doi.org/https://doi.org/10.1016/S0022-2836(03)00755-1

Van Etten, J.L., Agarkova, I. V, Dunigan, D.D., 2020. Chloroviruses. Viruses. https://doi.org/10.3390/v12010020

Vincent, F., Sheyn, U., Porat, Z., Schatz, D., Vardi, A., 2021. Visualizing active viral infection reveals diverse cell fates in synchronized algal bloom demise. Proc. Natl. Acad. Sci. 118, e2021586118. https://doi.org/10.1073/pnas.2021586118

Weitz, J.S., Stock, C.A., Wilhelm, S.W., Bourouiba, L., Coleman, M.L., Buchan, A., Follows, M.J., Fuhrman, J.A., Jover, L.F., Lennon, J.T., Middelboe, M., Sonderegger, D.L., Suttle, C.A., Taylor, B.P., Frede Thingstad, T., Wilson, W.H., Eric Wommack, K., 2015. A multitrophic model to quantify the effects of marine viruses on microbial food webs and ecosystem processes. ISME J. 9, 1352–1364. https://doi.org/10.1038/ismej.2014.220

Welsh, J.E., Steenhuis, P., de Moraes, K.R., van der Meer, J., Thieltges, D.W., Brussaard, C.P.D., 2020. Marine virus predation by non-host organisms. Sci. Rep. 10, 5221. https://doi.org/10.1038/s41598-020-61691-y

Wigington, C.H., Sonderegger, D., Brussaard, C.P.D., Buchan, A., Finke, J.F., Fuhrman, J.A., Lennon, J.T., Middelboe, M., Suttle, C.A., Stock, C., Wilson, W.H., Wommack, K.E., Wilhelm, S.W., Weitz, J.S., 2016. Re-examination of the relationship between marine virus and microbial cell abundances. Nat. Microbiol. 1, 15024. https://doi.org/10.1038/nmicrobiol.2015.24

Wilhelm, S.W., Suttle, C.A., 1999. Viruses and Nutrient Cycles in the Sea: Viruses play critical roles in the structure and function of aquatic food webs. Bioscience 49, 781–788. https://doi.org/10.2307/1313569

Wilhelm, S.W., Weinbauer, M.G., Suttle, C.A., Jeffrey, W.H., 1998. The role of sunlight in the removal and repair of viruses in the sea. Limnol. Oceanogr. 43, 586–592. https://doi.org/https://doi.org/10.4319/lo.1998.43.4.0586

Wilson, W.H., Schroeder, D.C., Allen, M.J., Holden, M.T.G., Parkhill, J., Barrell, B.G., Churcher, C., Hamlin, N., Mungall, K., Norbertczak, H., Quail, M.A., Price, C., Rabbinowitsch, E., Walker, D., Craigon, M., Roy, D., Ghazal, P., 2005. Complete Genome Sequence and Lytic Phase Transcription Profile of a Coccolithovirus. Science (80-.). 309, 1090–1092. https://doi.org/10.1126/science.1113109

Wilson, W.H., Tarran, G.A., Schroeder, D., Cox, M., Oke, J., Malin, G., 2002. Isolation of viruses responsible for the demise of an Emiliania huxleyi bloom in the English Channel. J. Mar. Biol. Assoc. United Kingdom 82, 369–377. https://doi.org/DOI: 10.1017/S002531540200560X

Xie, L., Zhang, R., Luo, Y.-W., 2022. Assessment of Explicit Representation of Dynamic Viral Processes in Regional Marine Ecological Models. Viruses. https://doi.org/10.3390/v14071448

Zemb, O., Achard, C.S., Hamelin, J., De Almeida, M.-L., Gabinaud, B., Cauquil, L., Verschuren, L.M.G., Godon, J.-J., 2020. Absolute quantitation of microbes using 16S rRNA gene metabarcoding: A rapid normalization of relative abundances by quantitative PCR targeting a 16S rRNA gene spike-in standard. Microbiologyopen 9, e977. https://doi.org/https://doi.org/10.1002/mbo3.977